

200 REPEATED ★ MIDTERMS CONCEPTUAL MCQs

COURSE

BT403/BT619 AGRICULTURE BIOTECHNOLOGY

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TO GET MAXIMUM MARKS

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BS BIOLOGICAL SCIENCES TEAM

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These MCQs are carefully selected from past midterms to help you understand key concepts.
Use this resource along with your official handouts for the best preparation and maximum results.
If any question appears in your exam, consider it a positive step towards your success!

Best of Luck!

JAWAD CHAUDHARY

MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN BT403/BT619 AGRICULTURE BIOTECHNOLOGY MIDTERM EXAMS

Main Group (BS Biological Sciences) https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN

BT403: Agricultural Biotechnology https://chat.whatsapp.com/Iq0lFe4dAU20sIGNdWjJmh

Q1. The practice of rapidly multiplying elite plant material through tissue culture techniques is called _____.

Answer: Micropropagation

Q2. The initial phase of micropropagation involving maintenance of donor plants under controlled conditions is known as _____.

Answer: Step 0

Q3. The stage of micropropagation primarily responsible for culture initiation and establishment is Stage _____.

Answer: I

Q4. Rapid multiplication of shoots mainly occurs during Stage _____ of micropropagation.

Answer: II

Q5. In vitro development of roots generally takes place during Stage _____.

Answer: III

Q6. Transfer of plantlets from laboratory to greenhouse conditions occurs during the _____ stage.

Answer: Acclimatization

Q7. Shoot tips and axillary shoots are commonly used as _____ in micropropagation.

Answer: Explants

Q8. Plants with vigorous germination and branching ability are generally more suitable for _____.

Answer: Micropropagation

Q9. Recently produced plant tissues are preferred as explants because they possess better _____ potential.

Answer: Regenerative

Q10. Growth regulators commonly modified during rooting stage include auxins and _____.

Answer: Cytokinins

Q11. A photoperiod of _____ hours light and 8 hours dark is generally suitable for shoot proliferation.

Answer: 16

Q12. The optimum temperature for most micropropagation cultures is around _____ °C.

Answer: 25

Q13. Production of disease-free plants is a major application of _____.

Answer: Micropropagation

Q14. Micropropagation is considered the only practical method for multiplying genetically modified cells after _____ fusion.

Answer: Protoplast

Q15. Synthetic seeds are commonly produced using _____ embryogenesis.

Answer: Somatic

Q16. Synthetic seeds are defined as artificially _____ propagules capable of developing into plants.

Answer: Encapsulated

Q17. Synthetic seeds functionally mimic natural _____.

Answer: Seeds

Q18. One major advantage of synthetic seeds is the maintenance of _____ uniformity.

Answer: Genetic

Q19. The direct field delivery of propagules without transplantation is an advantage of _____ seeds.

Answer: Synthetic

Q20. Germplasm refers to collections of living _____ resources used in breeding programs.

Answer: Genetic

Q21. Germplasm conservation helps preserve valuable plant _____.

Answer: Diversity

Q22. Variation observed among plants regenerated through tissue culture is called _____ variation.

Answer: Somaclonal

Q23. Variations arising from plants regenerated from gametic cells are termed _____ variation.

Answer: Gametoclonal

Q24. Chromosomal rearrangements are an important source of _____ variation.

Answer: Somaclonal

Q25. Development of pathogen-free planting material can be achieved through culture of _____ tissues.

Answer: Meristem

Q26. The process of improving crop plants through controlled mating and selection is called plant _____.

Answer: Breeding

Q27. Crossing between genetically distant species is known as _____ hybridization.

Answer: Wide

Q28. Incompatibility occurring before fertilization is called a _____-zygotic barrier.

Answer: Pre

Q29. Failure of embryo development after fertilization represents a _____-zygotic barrier.

Answer: Post

Q30. The in vitro technique used to rescue immature embryos from incompatible crosses is called embryo _____.

Answer: Rescue

Q31. Fusion of somatic cells without cell walls results in the formation of _____.

Answer: Protoplasts

Q32. The artificial fusion of protoplasts from different species is called protoplast _____.

Answer: Fusion

Q33. A hybrid cell produced through fusion of protoplasts from different species is called a _____.

Answer: Somatic hybrid

Q34. The partial transfer of genetic material through asymmetric fusion results in a _____.

Answer: Cybrid

Q35. The enzyme commonly used to degrade plant cell walls during protoplast isolation is _____.

Answer: Cellulase

Q36. Another enzyme used to digest the pectin layer during protoplast isolation is _____.

Answer: Pectinase

Q37. Successful plant regeneration from isolated protoplasts demonstrates cellular _____.

Answer: Totipotency

Q38. The ability of a single plant cell to regenerate into a complete plant is called _____.

Answer: Totipotency

Q39. A nutrient medium containing all essential mineral elements is necessary for plant tissue _____.

Answer: Culture

Q40. Murashige and Skoog medium is commonly abbreviated as _____ medium.

Answer: MS

Q41. Strict aseptic conditions are essential in tissue culture to prevent _____.

Answer: Contamination

Q42. The process of eliminating microorganisms from explants is called surface _____.

Answer: Sterilization

Q43. The enclosed workspace designed to maintain sterile conditions during culture transfer is called a laminar airflow _____.

Answer: Cabinet

Q44. Ethanol and sodium hypochlorite are commonly used as _____ agents.

Answer: Sterilizing

Q45. Unorganized masses of proliferating plant cells formed in vitro are called _____.

Answer: Callus

Q46. Direct regeneration of shoots from explants without callus formation is called direct _____.

Answer: Organogenesis

Q47. Formation of shoots and roots through an intermediate callus phase is called indirect _____.

Answer: Organogenesis

Q48. Development of embryos from somatic cells without fertilization is termed somatic _____.

Answer: Embryogenesis

Q49. Somatic embryos resemble zygotic embryos in their developmental _____.

Answer: Pattern

Q50. The ability of cultured tissues to produce complete plants depends primarily on cellular _____.

Answer: Totipotency

Q51. The introduction of foreign DNA into a plant genome is called genetic _____.

Answer: Transformation

Q52. The soil bacterium commonly used as a natural gene transfer vector in plants is _____.

Answer: Agrobacterium

Q53. Agrobacterium tumefaciens naturally causes _____ gall disease in plants.

Answer: Crown

Q54. The tumor-inducing plasmid present in Agrobacterium is called the _____ plasmid.

Answer: Ti

Q55. The DNA segment transferred from Agrobacterium into plant cells is known as _____-DNA.

Answer: T

Q56. Genes responsible for transferring T-DNA are called _____ genes.

Answer: Vir

Q57. The use of Agrobacterium for introducing desired genes into plants is called Agrobacterium-mediated _____.

Answer: Transformation

Q58. The direct introduction of DNA into plant cells using microscopic metal particles is called particle _____.

Answer: Bombardment

Q59. The gene gun method is also referred to as _____ transformation.

Answer: Biolistic

Q60. Gold and tungsten particles are commonly used in the _____ method.

Answer: Biolistic

Q61. A gene introduced into an organism from another species is called a _____ gene.

Answer: Transgene

Q62. Plants carrying and expressing foreign genes are called _____ plants.

Answer: Transgenic

Q63. Marker genes used to identify successfully transformed cells are called _____ markers.

Answer: Selectable

Q64. Antibiotic resistance genes are commonly used as _____ marker genes.

Answer: Selectable

Q65. The reporter gene frequently used to monitor gene expression in plants is the _____ gene.

Answer: GUS

Q66. The Green Fluorescent Protein reporter gene is abbreviated as _____.

Answer: GFP

Q67. Production of pharmaceutical compounds using genetically modified plants is known as molecular _____.

Answer: Farming

Q68. Plants used for producing vaccines and therapeutic proteins are often called plant _____.

Answer: Bioreactors

Q69. Edible vaccines are produced through _____ technology.

Answer: Transgenic

Q70. The use of living microorganisms to enhance plant growth is known as _____ fertilization.

Answer: Bio

Q71. Microorganisms that improve nutrient availability to plants are called _____ fertilizers.

Answer: Bio

Q72. Rhizobium is a biofertilizer mainly associated with biological _____ fixation.

Answer: Nitrogen

Q73. Blue-green algae contribute to soil fertility through atmospheric nitrogen _____.

Answer: Fixation

Q74. Microorganisms used to control agricultural pests are known as _____ pesticides.

Answer: Bio

Q75. Bacillus thuringiensis produces insecticidal proteins called _____ toxins.

Answer: Cry

Q76. Transgenic crops expressing Cry proteins for insect resistance are commonly known as _____ crops.

Answer: Bt

Q77. The insecticidal activity of Bt toxins is primarily due to the formation of pores in the insect _____.

Answer: Midgut

Q78. Bt toxins become activated under _____ conditions present in the insect gut.

Answer: Alkaline

Q79. The major advantage of Bt crops is the reduced dependence on chemical _____.

Answer: Insecticides

Q80. Development of resistance in insect populations can occur due to continuous exposure to Bt _____.

Answer: Toxins

Q81. The strategy of planting non-Bt crops alongside Bt crops to delay resistance development is called the _____ strategy.

Answer: Refuge

Q82. A pesticide is a substance used to prevent, destroy, repel, or control _____.

Answer: Pests

Q83. Pesticides specifically designed to control insect populations are called _____.

Answer: Insecticides

Q84. Chemicals used to manage unwanted plant species are known as _____.

Answer: Herbicides

Q85. Pesticides used to control fungal pathogens are called _____.

Answer: Fungicides

Q86. Rodenticides are pesticides developed to control _____.

Answer: Rodents

Q87. Organophosphate pesticides exert toxicity primarily by inhibiting the enzyme _____.

Answer: Acetylcholinesterase

Q88. Accumulation of acetylcholine at nerve endings results from _____ inhibition.

Answer: Acetylcholinesterase

Q89. Excessive salivation, sweating, and muscle twitching are characteristic signs of _____ pesticide poisoning.

Answer: Organophosphate

Q90. Integrated Pest Management emphasizes minimizing dependence on chemical _____.

Answer: Pesticides

Q91. Combining biological, cultural, and chemical methods for pest control is called Integrated Pest _____.

Answer: Management

Q92. The use of natural enemies to suppress pest populations is referred to as _____ control.

Answer: Biological

Q93. Ladybird beetles are commonly used as biological control agents against _____.

Answer: Aphids

Q94. Pesticide residues may accumulate through food chains in a process known as biological _____.

Answer: Magnification

Q95. Long-term persistence of pesticides in the environment can negatively affect _____ organisms.

Answer: Non-target

Q96. The immediate step in treating pesticide poisoning is to remove the patient from the source of _____.

Answer: Exposure

Q97. Removal of contaminated clothing helps reduce further pesticide _____.

Answer: Absorption

Q98. In cases of organophosphate poisoning, _____ is commonly administered as an antidote to counteract muscarinic effects.

Answer: Atropine

Q99. Pralidoxime (2-PAM) is used therapeutically to reactivate inhibited _____.

Answer: Acetylcholinesterase

Q100. Early recognition and prompt _____ are essential for successful management of pesticide poisoning.

Answer: Treatment

Q101. The branch of biotechnology that applies biological techniques for crop improvement is called agricultural _____.

Answer: Biotechnology

Q102. The ability of plants to regenerate from a single cell demonstrates cellular _____.

Answer: Totipotency

Q103. Tissue culture techniques are generally performed under _____ conditions.

Answer: Aseptic

Q104. The artificial nutrient medium used for plant growth in vitro contains macro and _____ nutrients.

Answer: Micro

Q105. Vitamins added to culture media function as growth _____.

Answer: Factors

Q106. Sucrose in tissue culture medium primarily serves as a carbon _____.

Answer: Source

Q107. Agar is commonly used in tissue culture media as a _____ agent.

Answer: Solidifying

Q108. The pH of most plant tissue culture media is adjusted before _____.

Answer: Autoclaving

Q109. High-pressure steam sterilization is performed using an _____.

Answer: Autoclave

Q110. Culture vessels should remain free from microbial _____.

Answer: Contamination

Q111. Dedifferentiation of mature plant cells often results in _____ formation.

Answer: Callus

Q112. The redifferentiation process allows callus cells to form organized _____.

Answer: Structures

Q113. Shoot formation from cultured tissues is called _____ morphogenesis.

Answer: Caulogenesis

Q114. Root formation from cultured tissues is termed _____.

Answer: Rhizogenesis

Q115. A higher cytokinin-to-auxin ratio generally favors _____ development.

Answer: Shoot

Q116. A higher auxin-to-cytokinin ratio generally promotes _____ formation.

Answer: Root

Q117. Organogenesis occurring without callus formation is called _____ organogenesis.

Answer: Direct

Q118. Organogenesis involving an intermediate callus stage is called _____ organogenesis.

Answer: Indirect

Q119. The process of developing embryos from non-reproductive cells is somatic _____.

Answer: Embryogenesis

Q120. Somatic embryos are bipolar structures capable of developing into complete _____.

Answer: Plants

Q121. Genetic variation observed among tissue culture-derived plants is called _____ variation.

Answer: Somaclonal

Q122. Somaclonal variation can result from chromosomal _____.

Answer: Rearrangements

Q123. Changes in chromosome number may contribute to somaclonal _____.

Answer: Variation

Q124. Somaclonal variation provides a source of novel genetic _____.

Answer: Variability

Q125. Disease-resistant plant variants can sometimes be selected from _____ cultures.

Answer: Callus

Q126. Preservation of valuable genetic resources is known as germplasm _____.

Answer: Conservation

Q127. Long-term storage of plant materials at ultra-low temperatures is called _____ preservation.

Answer: Cryo

Q128. Liquid nitrogen used in cryopreservation has a temperature of approximately _____ °C.

Answer: -196

Q129. Cryopreservation minimizes cellular metabolic _____.

Answer: Activity

Q130. Seed banks are important facilities for germplasm _____.

Answer: Storage

Q131. Plants possessing a single set of chromosomes are called _____.

Answer: Haploids

Q132. Haploid plants can be produced through _____ culture.

Answer: Anther

Q133. The culture of isolated pollen grains is called _____ culture.

Answer: Microspore

Q134. Haploid production accelerates the development of homozygous _____.

Answer: Lines

Q135. Chromosome doubling in haploids produces _____ haploids.

Answer: Doubled

Q136. Protoplasts are plant cells lacking a _____ wall.

Answer: Cell

Q137. Cell wall removal is achieved through enzymatic _____.

Answer: Digestion

Q138. Polyethylene glycol is commonly abbreviated as _____.

Answer: PEG

Q139. PEG is frequently used to induce protoplast _____.

Answer: Fusion

Q140. Somatic hybridization bypasses barriers associated with sexual _____.

Answer: Reproduction

Q141. The introduction of useful genes into crops aims at genetic _____.

Answer: Improvement

Q142. A DNA molecule used to transfer genes into host cells is called a _____.

Answer: Vector

Q143. Agrobacterium-mediated transformation utilizes a natural DNA transfer _____.

Answer: System

Q144. The Ti plasmid originally induces tumor formation in susceptible _____.

Answer: Plants

Q145. Disarmed Ti plasmids are modified to eliminate disease-causing _____.

Answer: Genes

Q146. Transformation efficiency refers to the frequency of successful gene _____.

Answer: Transfer

Q147. Reporter genes help monitor the expression of transferred _____.

Answer: Genes

Q148. Selectable markers aid in identifying transformed _____.

Answer: Cells

Q149. Herbicide resistance genes are commonly used as selectable _____.

Answer: Markers

Q150. Gene expression is influenced by regulatory DNA sequences called _____.

Answer: Promoters

Q151. Crops containing introduced foreign genes are called _____ crops.

Answer: Transgenic

Q152. Bt cotton was developed to enhance resistance against insect _____.

Answer: Pests

Q153. Cry proteins specifically target susceptible insect _____.

Answer: Larvae

Q154. Transgenic plants with virus resistance often express viral _____ proteins.

Answer: Coat

Q155. Nutritional enhancement through biotechnology is called _____ fortification.

Answer: Bio

Q156. Biofertilizers improve soil fertility through beneficial microbial _____.

Answer: Activity

Q157. Azotobacter is a free-living nitrogen-fixing _____.

Answer: Bacterium

Q158. Rhizobium forms symbiotic associations with _____ plants.

Answer: Leguminous

Q159. Phosphate-solubilizing microorganisms enhance phosphorus _____.

Answer: Availability

Q160. Cyanobacteria contribute to agricultural productivity by fixing atmospheric _____.

Answer: Nitrogen

Q161. Biopesticides are derived from natural biological _____.

Answer: Sources

Q162. Bacillus thuringiensis is widely used as a microbial _____.

Answer: Insecticide

Q163. Biopesticides are generally considered more environmentally _____.

Answer: Friendly

Q164. Biopesticides often exhibit specificity toward target _____.

Answer: Pests

Q165. Reduced environmental persistence is an advantage of _____ pesticides.

Answer: Bio

Q166. Excessive pesticide use may lead to pest _____ development.

Answer: Resistance

Q167. Persistent pesticides can accumulate in adipose _____.

Answer: Tissue

Q168. The concentration of toxic substances at higher trophic levels is called biological _____.

Answer: Magnification

Q169. Acute pesticide poisoning generally results from short-term _____ exposure.

Answer: High

Q170. Chronic pesticide toxicity develops following repeated low-dose _____.

Answer: Exposure

Q171. IPM stands for Integrated Pest _____.

Answer: Management

Q172. Economic threshold levels help determine the necessity of pest _____.

Answer: Control

Q173. Monitoring pest populations is an essential component of _____.

Answer: IPM

Q174. Crop rotation is considered a _____ control method.

Answer: Cultural

Q175. The use of predators and parasites represents _____ control.

Answer: Biological

Q176. Organophosphates primarily affect the _____ system.

Answer: Nervous

Q177. Inhibition of acetylcholinesterase causes accumulation of _____.

Answer: Acetylcholine

Q178. Constricted pupils are medically termed _____.

Answer: Miosis

Q179. Respiratory failure is a severe complication of pesticide _____.

Answer: Poisoning

Q180. Immediate medical intervention improves the prognosis of poisoned _____.

Answer: Patients

Q181. Decontamination aims to reduce continued pesticide _____.

Answer: Exposure

Q182. Skin exposed to pesticides should be washed thoroughly with _____.

Answer: Water

Q183. Gastric lavage may be considered under appropriate _____ supervision.

Answer: Medical

Q184. Atropine antagonizes excessive _____ stimulation.

Answer: Cholinergic

Q185. Pralidoxime therapy is most effective when administered _____.

Answer: Early

Q186. Supportive management includes maintenance of adequate _____.

Answer: Respiration

Q187. Oxygen supplementation may be required in severe poisoning _____.

Answer: Cases

Q188. Monitoring vital signs is essential during poisoning _____.

Answer: Management

Q189. Prevention remains the most effective strategy against pesticide _____.

Answer: Poisoning

Q190. Proper use of personal protective equipment minimizes occupational _____.

Answer: Exposure

Q191. Plant biotechnology contributes to sustainable agricultural _____.

Answer: Development

Q192. Disease-resistant varieties reduce dependence on chemical _____.

Answer: Pesticides

Q193. Tissue culture enables year-round production of planting _____.

Answer: Material

Q194. Genetic engineering can improve crop nutritional _____.

Answer: Quality

Q195. Biotechnology-based approaches support food _____.

Answer: Security

Q196. Improved crop varieties can contribute to higher agricultural _____.

Answer: Productivity

Q197. Conservation of genetic resources ensures future breeding _____.

Answer: Potential

Q198. Environmentally safer pest management practices promote ecological _____.

Answer: Balance

Q199. Biotechnology integrates principles from biology, agriculture, and _____.

Answer: Technology

Q200. Responsible application of biotechnology supports sustainable _____.

Answer: Agriculture

🚨 200 Most Important Repeated Conceptual MCQs 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

BS Biological Sciences Team 🚀🎓

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